

# FENGRUN LIU

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## EDUCATION

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|-------------------------------------------------------------------------|------------------------|
| <b>Carnegie Mellon University (CMU)</b>                                 | PA, USA                |
| Ph.D. in Computer Science; Advisor: Prof. <a href="#">Wenting Zheng</a> | 2025 – 2030 (Expected) |
| <b>University of Science and Technology of China (USTC)</b>             | Anhui, China           |
| M.Eng. in Cyberspace Security                                           | 2022 – 2025            |
| <b>University of Electronic Science and Technology of China (UESTC)</b> | Chengdu, China         |
| B.Eng. in Software Engineering                                          | 2018 – 2022            |

## PUBLICATIONS

- [1] **HasteBoots: Proving TFHE Programmable Bootstrapping in Seconds.**  
Fengrun Liu\*, Hao-fei Liang, Xiang Xie, Yu Yu, Wenting Zheng, Yuncong Hu.  
*USENIX Security 2026* [\[pdf\]](#) [\[code\]](#)
- [2] **Scalable Multi-Party Computation Protocols for Machine Learning in the Honest-Majority Setting.**  
Fengrun Liu\*, Xiang Xie, Yu Yu.  
*USENIX Security 2024* [\[pdf\]](#) [\[code\]](#)
- [3] **Akita: A High-Performance Lattice-Based Polynomial Commitment Scheme.**  
Quang Dao, Omid Bodaghi, Amirhossein Khajepour, Giuseppe Vitto, Mohammadtaghi Badakhshan, Markos Georgiades, **Fengrun Liu\***, Jiapeng Zhang, Justin Thaler.  
*IACR ePrint 2026/1983* [\[eprint\]](#) [\[code\]](#)
- [4] **Edabits Generation with Quasi-Linear Communication Complexity.**  
Alexander Bienstock, Daniel Escudero, Qijia Fan, **Fengrun Liu\***, Yifan Song.  
*In submission; authors listed alphabetically.*

## WORK EXPERIENCE

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                 |
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| <b>LayerZero</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Canada (Remote) |
| Cryptography Intern                                                                                                                                                                                                                                                                                                                                                                                                                                                    | May – Aug. 2026 |
| <ul style="list-style-type: none"><li>Contributed to the design and development of <b>Akita</b>, a high-performance lattice-based polynomial commitment scheme, with a focus on GPU-friendly protocol design. <a href="#">[paper]</a></li><li>Implemented engineering optimizations to improve verifier performance in the <a href="#">Akita codebase</a>.</li><li>Authored the <a href="#">Akita Book</a>, documenting the protocol and its implementation.</li></ul> |                 |

## RESEARCH EXPERIENCE

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| <b>HasteBoots: Publicly Verifiable TFHE Programmable Bootstrapping [1]</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2024–2025 |
| Supervised by Prof. <a href="#">Yuncong Hu</a> and Prof. <a href="#">Wenting Zheng</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |           |
| <ul style="list-style-type: none"><li>Led protocol design and implementation of HasteBoots, a succinct proof system for verifying the correctness of outsourced TFHE computation with programmable bootstrapping.</li><li>Developed PIOPs for negacyclic NTT and batched arithmetic over quotient rings with <b>linear-time proving and sublinear verification</b>; decomposed blind rotation into batched relations and reduced non-arithmetic operations to lookups.</li><li>Implemented the system in <b>Rust</b>; with Dory commitments, proved a TFHE evaluation with bootstrapping in <b>about 10 seconds</b> and batches of <b>16 within one minute</b>, with 126 ms batch verification.</li></ul> |           |
| <b>Scalable MPC Protocols for Privacy-Preserving Machine Learning [2]</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2023      |
| Supervised by Prof. <a href="#">Yu Yu</a> and Dr. <a href="#">Xiang Xie</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |           |
| <ul style="list-style-type: none"><li>Led protocol design and implementation of a scalable framework for private neural-network inference with semi-honest security in the honest-majority setting.</li><li>Designed Mersenne-prime truncation with a <b>1-bit gap</b> between shares and secrets, and <b>one-round online</b> fixed-point multiplication and bitwise comparison; developed efficient ReLU and MaxPool protocols.</li><li>Implemented and evaluated a <b>15.4k-line C++</b> framework across <b>3–63 parties</b>, demonstrating scalable private inference under LAN and WAN settings.</li></ul>                                                                                          |           |